# Spatial Analysis

April 19, 2023

## 1 Methods

For each sample, we will compare the distribution of each markers in the normal and lesion regions.

- Markers for DD: CD3, CD4, CD8a, CD11c, CD15, CD45, CD68, CD69, CD123, CD279
- Markers for CF: CD1a, CD3, CD4, CD8a, CD11c, CD15, CD45, CD69, CD123, CD279 (CD68 did not pass QC)

try flow 0.5 or 0.6 to get the large bright ones
also find a way to filter out masks in the debris area, where one cell may disintegrate and each part may be called a cell, maybe by intensity threshold.

compare distance between peaks with the width of lesion, as measured by a smooth outline drawn by K

## 2 Results

| | Counts | | | | Counts/mm2 | | | |
|---|---|---|---|---|---|---|---|---|
| | normal | | lesion | | normal | | lesion | |
| | epi | dermis | epi | dermis | epi | dermis | epi | dermis |
| CD3 | 74 | 1222 | 158 | 5395 | 631 | 2874 | 121 | 1399 |
| CD15 | 11 | 238 | 6567 | 2867 | 93 | 559 | 5050 | 743 |
| CD123 | 131 | 879 | 1350 | 8038 | 1117 | 2067 | 1038 | 2084 |
| CD11c | 343 | 2056 | 8029 | 17905 | 2926 | 4836 | 6174 | 4643 |
| CD68 | 154 | 998 | 2524 | 11081 | 1314 | 2347 | 1941 | 2874 |
| area | 0.12 | 0.43 | 1.30 | 3.86 | | | | |

Table 1: DD

| | Counts | | | | Counts/mm2 | | | |
|---|---|---|---|---|---|---|---|---|
| | normal | | lesion | | normal | | lesion | |
| | epi | dermis | epi | dermis | epi | dermis | epi | dermis |
| CD3 | 422 | 8435 | 331 | 5629 | 976 | 4476 | 819 | 5574 |
| CD15 | 7 | 2154 | 756 | 391 | 16 | 1143 | 1871 | 387 |
| CD123 | 76 | 3680 | 324 | 1298 | 175 | 1952 | 801 | 1285 |
| CD11c | 624 | 8498 | 1308 | 4299 | 1443 | 4509 | 3237 | 4257 |
| area | 0.43 | 1.88 | 0.40 | 1.01 | | | | |

Table 2: CF

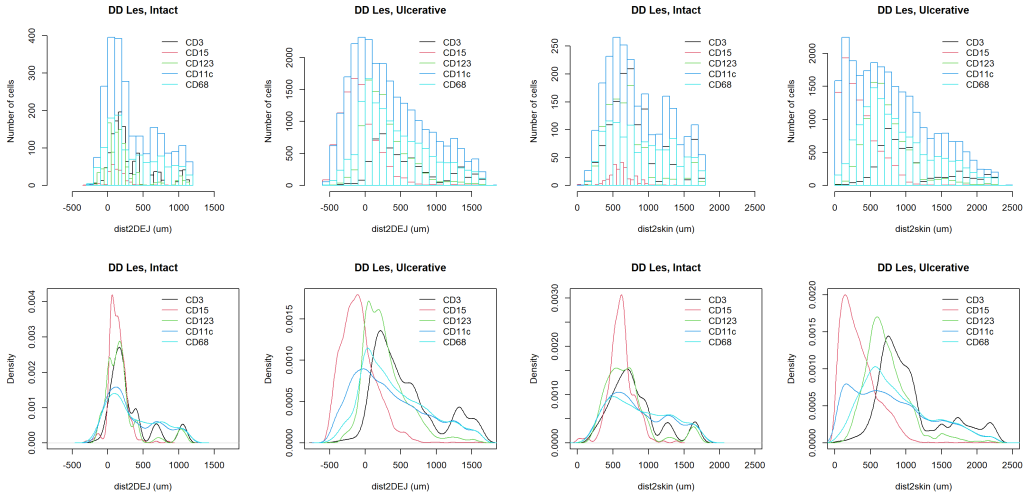



Figure 1: Distance to DEJ (left) and skin surface(right).

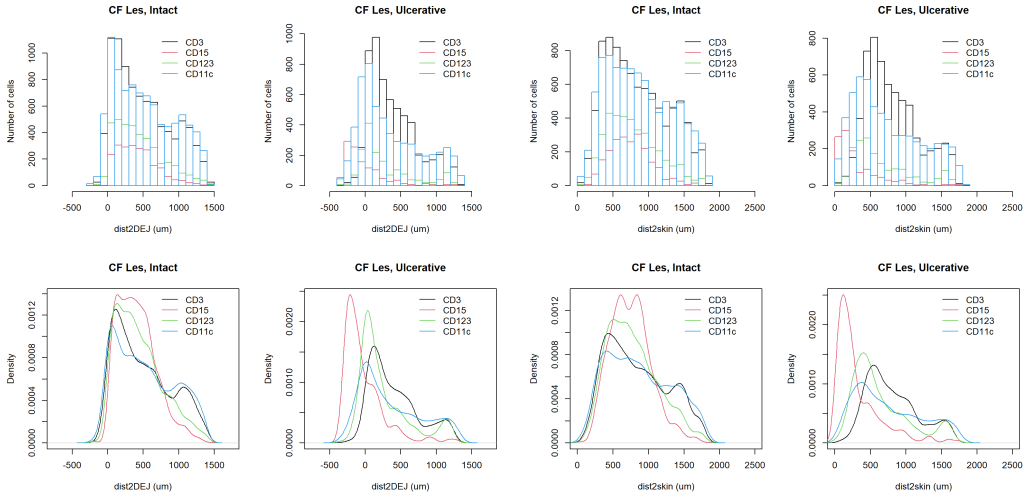



Figure 2: Distance to DEJ (left) and skin surface(right).